

Interconnection Network Architectures

1 Overview

Parallel database systems rely on high-speed interconnection networks to facilitate communication between processors and disks. The choice of topology significantly impacts the **scalability** and **communication delay** of the system.

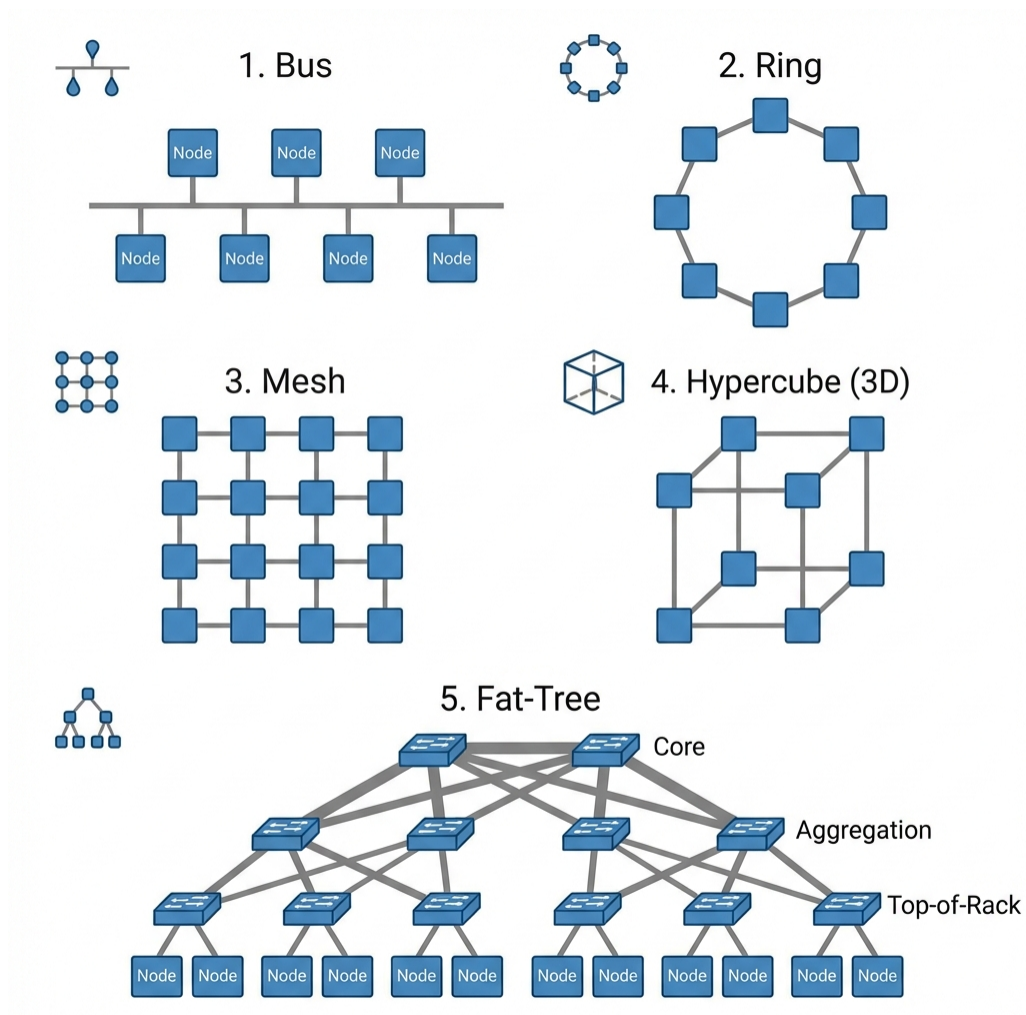


Figure 1: Common Interconnection Network Topologies.

2 Traditional Topologies

2.1 Bus Architecture

All system components send and receive data via a single, shared communication bus.

- **Characteristic:** Simple to implement.

- **Limitation:** Does not scale well as the number of processors increases; the bus becomes a massive bottleneck.

2.2 Ring Architecture

Components are connected in a closed-loop or ring structure. Each node is connected to exactly two others.

- **Characteristic:** Data travels in one or two directions around the ring.
- **Limitation:** A failure in one node or link can potentially disrupt the entire network (if not designed with redundancy).

2.3 Mesh Architecture

Components are arranged in a 2D grid/mesh, where each node is connected to its adjacent neighbors.

- **Scalability:** Better than a bus, as communication links grow with the number of components.
- **Delay:** Highly variable; may require $2\sqrt{n}$ hops (or $\sqrt{n}$ with wraparound) to traverse the grid.

2.4 Hypercube Architecture

Nodes are numbered in binary and connected if their binary representations differ by exactly one bit.

- **Performance:** n components can reach each other in at most $\log(n)$ hops.
- **Benefit:** Significantly reduces communication delays compared to mesh architectures.

3 Modern Data Center Architectures

Data centers today predominantly use hierarchical, tree-like topologies to manage hundreds of thousands of servers.

Tree-like / Fat-Tree Topology

A high-bandwidth network fabric structured in layers:

- **Top-of-Rack (ToR) Switch:** Connects approximately 40 machines within a single rack.
- **Aggregation Switches:** Connect multiple ToR switches.
- **Core Switches:** Connect aggregation layers together to form a massive **Data Center Fabric**.

4 Network Technologies

The choice of physical networking protocol determines the throughput and latency of the parallel system.

Technology	Typical Bandwidth	Latency Characteristics
Ethernet	1–100 Gbps	Several microseconds (μs)
Fiber Channel	32–138 Gbps	Optimized for storage area networks
Infiniband	High-performance tiers	Extremely Low (0.5–0.7 μs)

Infiniband vs. Ethernet

While Ethernet is common and cost-effective, **Infiniband** is the gold standard for parallel processing due to its sub-microsecond latency, which is essential for massive scale-out synchronization.